

UNIT 4

Fun with science

Science is a way of life. Science is a perspective. Science is the process that takes us from confusion to understanding in a manner that's precise, predictive and reliable ...

—Brian Greene



In this unit, you are going to:

- read a blog post about the human microbiome;
- write an article about friction;
- read a science magazine article about bionics;
- conduct a scientific experiment.

Welcome to the unit

Science provides explanations for many everyday occurrences. Read the leaflet below and discuss the following questions in pairs.

Why do stars twinkle?

When a star shines its steady light down on us through our atmosphere, the air bends the light. As the air moves, its temperature and density change, affecting the way the light bends. Therefore, from our perspective, the starlight changes slightly in brightness and position. This makes the star seem to twinkle. Between us and the stars closer to the horizon, there is a lot more air, so they appear to twinkle more than those stars higher in the sky.



Why are flowers fragrant?

The fragrance of a flower is released by its tissue. Each fragrance comes from a mixture of chemicals that float in the air. Each flower's fragrance is unique, attracting particular insects to the flower, which help spread its pollen so that seeds can be produced. Therefore, a flower's fragrance is essential for the plant's survival.

- 1 What causes stars to twinkle and flowers to be fragrant? Explain in your own words.
- 2 What other fun scientific facts do you know?

Reading



The world we live in is filled with microorganisms. Below is a blog post by a microbiologist describing these tiny life forms. Before you read the blog post, think about the following questions:

- Where can you find microorganisms?
- How do you think microorganisms harm or benefit us?

The human microbiome: an invisible microuniverse



What organisms live with us, make up about two per cent of an adult's body mass, yet are invisible to the naked eye? They are the estimated 39 trillion microorganisms living in and on our body, like bacteria and viruses, which compose the human microbiome! But there is no need to panic, for most of them are there to keep us healthy. Even though
5 incredibly small, these microorganisms influence our body and mind in big unexpected ways.

So, where do these microorganisms come from? Well, they have always been a part of us. Our very first birthday gift—three quarters of the essential microorganisms we need to survive and grow—is from our mother when we are born. By the time we reach the
10 age of two, we have our very own personal collection of microorganisms that reflects the various circumstances of our lives. Going through life, we further expose ourselves to foreign microorganisms, which cause a part of our microbiome to change every day. However, our core set of microorganisms, about two thirds, stays fairly constant throughout our lives.

15 The make-up and state of our special blend of microorganisms affects us physically, including our immune system, weight and sleep patterns. Every day, the resident microorganisms in our body compete with the invading ones, preventing them from getting a foothold, thus helping form a strong immune system and keep us healthy. Similarly, the composition of our gut microbiome plays a vital role in regulating our
20 body weight. Besides helping digest food, evidence indicates that gut microorganisms affect the way we store fat and respond to chemicals that make us feel hungry or full. As a result, without the proper mix of microorganisms in the gut, we might suffer from

health problems like obesity. Added to this, microorganisms can influence certain aspects of our sleep patterns. If we stay up all night or travel rapidly through many time zones, we may interrupt the microorganisms' biological schedules, which is likely to cause a disturbance in our normal sleep routine.

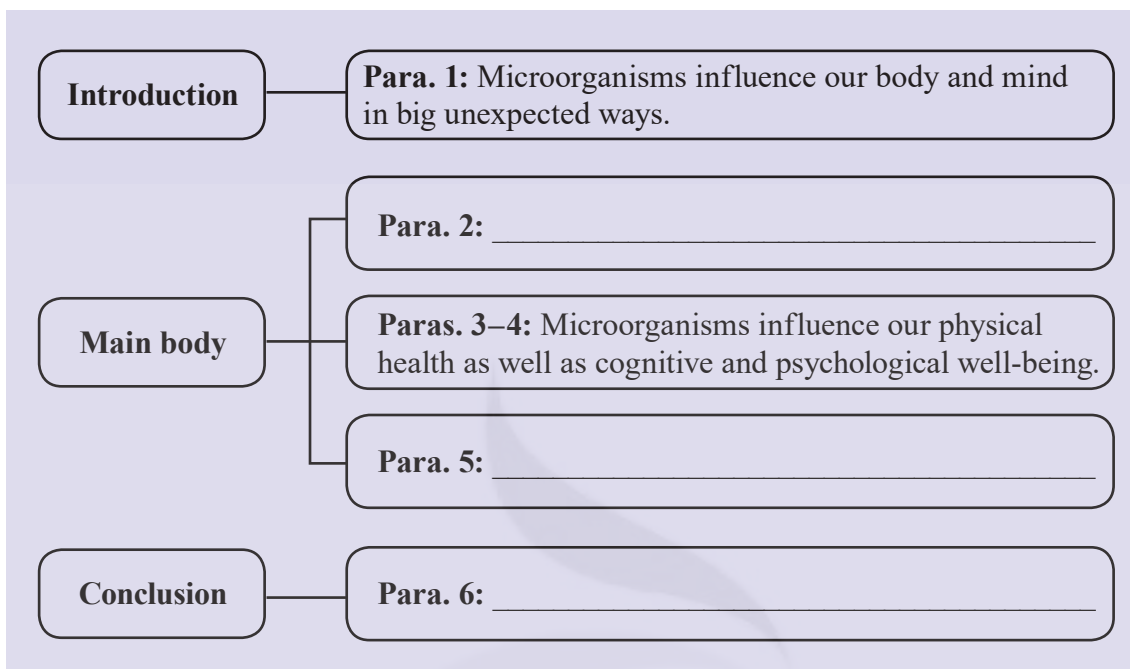
The influence of microorganisms goes beyond physical health to our cognitive and psychological well-being. Believe it or not, gut microorganisms affect how well the brain functions. While scientists are still at the early stages of uncovering exactly how gut microorganisms communicate with the brain, they do know that the brain and the gut “talk” to each other. For example, when the body is stressed, ill or old, some gut microorganisms respond negatively, releasing harmful substances which may enter the brain through the bloodstream. When the immune system attacks these substances, it often kills healthy brain nerve cells. Furthermore, gut microorganisms help regulate brain chemistry with certain chemicals they produce, so an imbalance of beneficial and harmful microorganisms in the gut can negatively affect brain chemistry. These gut-brain interactions sometimes result in such problems as memory loss and depression. It is therefore important that we take good care of our gut microorganisms so as to improve our cognitive capabilities and enhance our psychological health.

Given their importance, scientists are working hard to find out how we can take advantage of microorganisms within the field of human health. Successful projects have found ways to improve the immune system through microbiome transplants—using “good” microorganisms to replace “bad” ones. Furthermore, the findings of how the microbiome affects psychological health have given rise to a whole new class of medicines that could be used for treating serious mental illnesses, as well as relieving daily stress and anxiety. In the future, we might even be capable of developing “microbiome vaccines” that could get rid of current health issues like obesity and malnutrition.

Microorganisms, those tiny creatures we carry around all our lives, have surprised us with their seemingly limitless potential. More of their secrets will certainly be revealed as researchers continue to explore their world. The invisible microuniverse is filled with endless possibilities waiting to be explored, just like the great universe out there.

A Understanding the text

A1 Read the blog post and complete the chart below with the main idea of each paragraph.



A2 Read the blog post again carefully and answer the following questions.

1 How many microorganisms live in and on our body?

2 Why are the microorganisms we receive from our mother essential?

3 How does the gut microbiome regulate our body weight?

4 When will gut microorganisms release harmful substances?

5 What benefits could “microbiome vaccines” provide?

A3 In pairs, discuss the following questions.

1 How do you understand “microuniverse”? Why do you think the author compares the world of microorganisms to a tiny universe?

2 The author describes the big influences microorganisms have on our body and mind. What other encounters with microorganisms do we have in our daily lives?

B Building your language

B1 The passage below is about some important discoveries in the world of microorganisms. Complete the passage with the correct forms of the words and phrases in the box below.

reveal
incredibly

blend
imbalance

estimate
enhance

be capable of
the naked eye

It (1) _____ that our body, both inside and out, is home to trillions of microorganisms. However, because they are invisible to (2) _____, for most of human history, people did not even know that microorganisms existed. Then how did we learn about these (3) _____ small creatures? Let's journey back to the 1670s, when Dutchman Antonie van Leeuwenhoek opened a window into the hidden world of microorganisms. Studying a water droplet under his microscope, Van Leeuwenhoek observed "little animals" swimming and jumping in it. Thus, he became the first person to see microorganisms.

Another secret of microorganisms (4) _____ in the 1800s, when German physician Robert Koch linked them with disease. Suspecting that microorganisms caused some sheep to be sick, he grew the microorganisms in isolation from all living things. A mouse died after it was injected with the microorganisms. This confirmed the relationship between microorganisms and disease, which considerably (5) _____ people's understanding of microorganisms.

In the 20th century, American microbiologist David Relman discovered new microorganisms when he and his team sequenced the DNA of the bacteria found on his teeth. For the first time, people realized that the human body might be packed with a complex (6) _____ of previously undiscovered microorganisms.

Today ongoing research casts new light on the hidden world of microorganisms. Scientists are finding out that microorganisms (7) _____ regulating all aspects of our well-being. Indeed, a(n) (8) _____ of microorganisms in our body might seriously affect our lives.

B2 The blog post asks and answers a question to introduce the topic of the human microbiome. Find the question and answer in the blog post and use the same technique to introduce the topic of the applications of microorganisms in daily life. Use the following information to help you.

Applications of microorganisms in daily life	
Food and drinks	yoghurt, cheese, beer, wine, etc.
Medicine	probiotic drugs, antibiotics, etc.

Grammar and composition

Subjunctive mood (II): other cases

A Exploring the rules

Below is a story from a popular science magazine about Wilson Bentley, the “Snowflake Man”. Find the sentences that use the subjunctive mood and fill in the box below. The first one has been done for you.

We might all know the saying “no two snowflakes are alike”, but how many of us have heard of the “Snowflake Man”, Wilson Bentley, the person who actually made this discovery? It is important that he be remembered for revolutionizing our understanding of these amazing crystal structures.

Born in the snowy US state of Vermont in 1865, Bentley developed a strong fascination with snowflakes. He wished that he could capture their beauty on film. However, creating photos with fine details of each individual snowflake was by no means easy. After many failed attempts, he attached his camera to a microscope and set up his equipment outside, otherwise the snowflakes would have melted quickly. He worked in the freezing cold for hours at a time, carefully handling each snowflake with a feather, and photographing it with a 90-second exposure. Triumph was his when, in 1885, he successfully captured the world’s first image of a single snowflake!

Over the decades, Bentley photographed more than 5,000 snowflakes. Through many years of trial and error, he explored the secrets hidden from human eyes, as though he were digging up buried treasure. Eventually he discovered that no two snowflakes are ever exactly the same. Excited by what he saw, he stated, “Under the microscope, I found that snowflakes were miracles of beauty ... Every crystal was a masterpiece of design and no one design was ever repeated.” In the hope of sharing his discovery with the world, Bentley wrote the book *Snow Crystals*, which featured over 2,400 breathtaking photographs of snowflakes.

But for Bentley’s tireless work, we would not know that every single snowflake is unique. We have the “Snowflake Man” to thank for this amazing insight into the natural world.

It is important that he be remembered for ...

Working out the rules

- In place of *if*-clauses in implied conditionals, we can use *otherwise*, *without* and (1) _____.
- We can express unreal situations after *wish* and *would rather*.
- In *it is important/necessary/essential that ...*, we usually use “*should* + bare infinitive” in the *that*-clause. We can leave out (2) _____.
- We may use the subjunctive mood in clauses introduced by *as if*, *as though*, *even if* and *even though*.

➡ Grammar and composition notes → pages 94–95

B Applying the rules

B1 *There is one mistake in each of the sentences below. Circle the mistakes and correct them in the blanks. Make sure each sentence uses the subjunctive mood.*

1 Without the help of scientists, we are having a limited understanding of our world.

2 If only we have a better method of studying microorganisms.

3 The ten-year-old boy acts as though he is an expert in the field of physics.

4 The Chemistry teacher suggested that we conducted our own experiments.

5 I wish there is a photograph of every snowflake that has ever fallen to the ground!

B2 *Below is another article from the popular science magazine. Complete the article with the correct forms of the verbs in the brackets.*

Vast amounts of snow fall every year. However, even if you ⁽¹⁾ _____ (examine) trillions of snowflakes, you would not find two identical ones. In order to find out why each snowflake is unique, it is essential that we ⁽²⁾ _____ (understand) how they are formed.

Without the dust particles that float around in the Earth's atmosphere, perhaps snowflakes ⁽³⁾ _____ (not exist). When water vapour freezes around particles of dust in a cloud, ice crystals start to form, and this is the beginning of a snowflake. The water molecules in an ice crystal always organize themselves into six-sided shapes. As they move through the cloud, the "baby" snowflakes accumulate more and more water vapour. Once they are heavy enough, they start falling towards the ground.

During their downward journey through the Earth's atmosphere, the shapes of snowflakes are further influenced by temperature and humidity. Since each snowflake takes a different route, it encounters a unique mix of atmospheric conditions. Otherwise it ⁽⁴⁾ _____ (be) impossible for snowflakes to take on such an endless variety of shapes, such as needles, branches or even discs.

A "baby" snowflake moves along through the atmosphere on its way down, and continues to grow into a fully-formed "adult" snowflake before landing and disappearing. If only there ⁽⁵⁾ _____ (be) a way to track its remarkable journey! Wouldn't that be amazing?

B3 *Anna is talking to her friend Sophie about snowflakes. In pairs, complete the conversation below using the subjunctive mood.*

Anna: I'm impressed by these photographs of snowflakes. I only wish that ⁽¹⁾ _____.

Sophie: I would like to learn more about the science behind snowflakes. My Science teacher suggested that I ⁽²⁾ _____.

Anna: That's a great idea! But for ⁽³⁾ _____.

Integrated skills

Talking about friction

A Alison is reading a physics lecture transcript about friction. Read the lecture transcript below and answer the following questions.

Friction: a force that's hard to resist

What happens if you roll a ball across the grass and then across a smooth floor? Which surface lets you roll the ball faster and for longer? The answer may appear simple, but the reason behind it—friction—is very complicated. Defined as the force that makes it difficult for one object to slide over the surface of another or to move through a liquid or gas, friction has been fascinating scientists for centuries, and it's hard to resist!

In the late 1400s, Leonardo da Vinci—yes, the famous Italian artist—experimented with friction. He slid objects across different types of surfaces and made the connection between friction amount and its determining factors such as surface texture, even with the limited technological knowledge available then. He never published his discoveries, but he did apply this knowledge to his designs for various tools and machine parts.

Friction continued to draw scientists' attention over the centuries. In the 17th century, French physicist Guillaume Amontons drafted laws about it, which became the first laws of friction to be recorded in print. In the 20th century, Australian physicist Frank Philip Bowden investigated friction's relationship with lubrication. The 21st century has witnessed yet another milestone: friction may have a kind of “memory” that builds over time between two surfaces. This means friction depends on the history of contact between two surfaces, leading scientists to believe that it is a force more complex than previously thought. The finding may be of great importance as it could expand our understanding of physical systems that depend on friction, from micromachines to earthquakes.

The force of friction had been part of our lives long before we even knew that it existed. Today we recognize and value the necessary role it plays. As there is still much to learn about friction, who knows what the future holds for this essential force? What is certain, however, is that physicists are excited at its huge potential.

1 How do you understand that friction is a force that is “hard to resist”?

2 What important events in the study of friction are mentioned in the transcript?

B Alison is listening to an interview about friction. Listen and finish the exercises below.



B1 Listen to the interview and choose the best answer. Circle the letters of the correct options.

- 1 What is Ms Li's opinion about friction?
 - a Friction is generated by air particles.
 - b Friction enables us to go about our daily lives.
 - c Shooting stars enter the Earth's surface because of friction.
- 2 How can mechanical systems last longer?
 - a Generating heat.
 - b Rubbing moving parts together.
 - c Applying oil between the surfaces.
- 3 Which of the following experience the least air resistance?
 - a Hyperloop trains.
 - b Maglev trains.
 - c Shooting stars.
- 4 How does the man speaker feel at the end of the interview?
 - a Amused.
 - b Shocked.
 - c Excited.



B2 Listen to the interview again and complete the table below.

	Presence of friction	Decrease in or absence of friction
Present	- Allowing our fingers to hold (1) _____, preventing them from falling (2) _____ shooting stars before they reach the ground	Reducing (3) _____ in mechanical systems
Future	Producing an electric charge that could be used as an alternative source of power for (4) _____	Enabling hyperloop trains to travel through (5) _____ at speeds of over 1,000 km per hour

C In pairs, have a discussion about friction. Use the following questions to help you.

- What is friction?
- In what situations is friction useful?
- In what situations is friction unnecessary?

Expressions

Expressing certainty

It goes without saying that ...

Without doubt, ...

It is certain/clear/obvious that ...

Clearly/Obviously/Surely, ...

There's no doubt that ...

I'm sure/certain that ...

D Write an article about friction. Use your ideas from part C and the information in parts A and B to help you. Check and improve your writing after you finish it.

Tip

Using examples to explain unfamiliar concepts

When you are writing about a concept with which readers may not be familiar, it is a good idea to use everyday examples to help explain the concept. This helps readers relate to the concept you are describing.

Introduction
to friction

Title: _____

The benefits
of the
presence or
absence of
friction

Conclusion

Extended reading



Read the article from a science magazine about the field of bionics.



Humans are curious by nature, and about nature. Perfected over billions of years, nature's forms and functions have inspired us to imitate it and develop materials and technologies that improve our lives. Consequently, a science known as "bionics" was born. Although the term "bionics" officially dates back to 1958, the practice is centuries
5 old. It could be said that as far back as 2,500 years ago, Lu Ban, a carpenter in ancient China, was engaged in bionics when he produced a saw by copying the shape of a leaf's sharp teeth. Currently, bionics has scientists eagerly combing the natural world for original ideas to stimulate their creativity.

The idea of taking inspiration from nature may sound simple; however, scientists often
10 need to go to great lengths to effectively combine nature and innovation. Firstly, they must observe a living organism with the potential for offering solutions to a particular human problem. Secondly, during observation and data analysis, scientists should try to identify nature's own solutions, which may be used in design and engineering. Thirdly, further exploration is required to determine how the natural solution will
15 fit in with human needs. Finally, they need to decide whether the solution can be recreated to develop a new product or perhaps improve an existing one, and how this can be achieved. This process may have to be performed several times and it may take scientists years of trial and error to perfect the product.

Once scientists bridge the gap between nature and innovation, a path opens up to
20 many novel ideas. One idea grew from observing the sunflower's spiral seed pattern, which is not just eye-catching, but also intelligent. The seeds grow at the "golden angle" of 137.5° , ensuring all of them receive sunlight. To capture maximum sunlight, the sunflower's face also follows the sun's path. As capturing sunlight is the purpose of solar power plants, it is not surprising that engineers have decided to imitate the

25 sunflower's natural solutions. Solar mirrors are arranged to copy the spiral seed pattern and positioned at the same "golden angle". The mirrors are also programmed to follow the sun. Hence the solar power plant functions exactly like a sunflower. The solar power plant is also eco-friendly, requiring less land to build on and less energy to operate.

Equally exciting are the diverse possibilities provided by the animal kingdom. For
30 example, research into the brilliant blue wings of a species of butterfly has led to innovative technologies since scientists uncovered the secret behind their bright colour. The wings of these butterflies are covered with tiny diamond-shaped structures that interact with light, causing some colour wavelengths to be cancelled out while intensifying and reflecting others. Inspired, scientists have recreated these microstructures
35 that can be inset into legal documents and banknotes and produce a unique visual effect almost impossible to copy. A better understanding of the butterfly's vividly coloured wings has also led scientists to develop paints without using harmful substances, which not only last longer but are also more eco-friendly.

As we continue to study and appreciate the wonder of nature, it is certain that our
40 future will see many more fascinating technological innovations. Bionics is set to move forward quickly with potential growth in the areas of vision, hearing, artificial limbs and wearable robotics. For example, inspired by the incredibly strong exoskeleton of insects, scientists and engineers are now building robotic suits. When worn, these exoskeletons can greatly improve the mobility, strength and load-bearing capabilities of
45 the wearer. However, what we are witnessing today is just the tip of the iceberg. Once such obstacles as power supply, weight and range of motion are overcome, exoskeletons will have even wider applications. Indeed, a real-life Iron Man could be waiting for us in the future!

With such rapid advancements being made in bionics, we have come to realize how
50 clever nature is and how much it can show us. As we work hard to make life better for everyone, it is possible that nature has the answers to our most urgent questions, and all we have to do is look for them. Nature not only provides our lives with a scenic backdrop, it could also prove to be our greatest teacher!

A *Divide the article into different sections and summarize the main idea of each section. Explain your reasons for dividing the article in this way.*

B *What examples of bionics can you think of that have improved our lives?*

C *Think of something amazing in nature. How can it inspire an innovation that will benefit us?*

Project

Conducting a scientific experiment

A As a class, discuss different types of scientific experiments. You can use the ideas below or think of other ideas. Then in groups, choose one specific scientific experiment to research.

light	sound	magnetism
gravity	melting point	air pressure

Tip

Choosing a scientific experiment

When you choose a scientific experiment, consider the following factors:

- safety—avoid experiments that may cause danger, e.g. those about fire or explosion;
- workability—make sure that all the necessary equipment and materials are easily available, and avoid experiments that involve too many steps or take too much time.

B As a group, research your scientific experiment. Use the ideas below to help you.

- Purpose
- Materials needed
- Steps
- Dos and don'ts
- Principle
- Connections with everyday life

C As a group, conduct your scientific experiment. You can take photos or make a video of how you do the experiment. Use the example below to help you. Then present your experiment to the rest of the class.

Ice hook

Purpose

To explore how salt affects the freezing point of water

Materials needed

A piece of string, a bowl of cold water, salt, two ice cubes

Steps

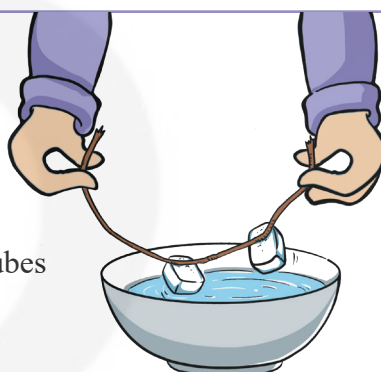
- 1 Put the ice cubes in the water.
- 2 Place the piece of string over the surface of the ice cubes.
- 3 Put a small amount of salt on the area where the string touches the ice cubes.
- 4 Wait for about 30 seconds.
- 5 Gently lift the ice cubes out of the water with the string. Then you will find the ice cubes attach themselves to the string as if hooked by it.

Principle

Adding salt makes water's freezing point lower, so the surface of the ice cubes melts faster. However, since only a small amount of salt is added, the rest of the ice is still frozen, and the melted ice soon refreezes. As a result, the string on the ice cubes' surface becomes attached to them.

Connections with everyday life

- 1 Salt can be used to melt ice on the road.
- 2 Salt can help form ice cream correctly.



Assessment

Complete the following learning journal to assess your performance. Then work in groups and exchange your learning journal with your partners.

Learning journal

In this unit I learnt _____

I was most interested in _____

I did well in _____

However, there is something I need to improve on: _____

In order to make more progress, I will _____

* Assess your learning of vocabulary and grammar by doing language practice on pages 72–73.

Further study



Science does not just happen in a lab; it is also a part of everyday life. Many things we do on a daily basis are associated with science. Search online to learn how knowledge from science subjects like Maths, Physics and Chemistry is applied in everyday life.



Science is not only important, but also fun! Filled with interesting facts, science opens our eyes and minds to the world around us, and even within us. Read a book to learn some fun facts about science and get inspired.





Wordlist 1

(by unit)

说明: 本表收录各单元阅读文章和语篇练习中所包含的生词和短语, 不收录指示语和讲解文字中所包含的生词和短语。带 * 的词条为《普通高中英语课程标准 (2017 年版 2020 年修订)》规定之外的词汇。

Unit 1

- *probe** /prəʊb/ *n.* (不载人) 空间探测器; 探究; 探测仪 (2)
- *geological** /ˌdʒi:ə'lɒdʒɪkl/ (*also geologic*) *adj.* 地质的 (2)
- formation** /fɔ:'meɪʃn/ *n.* 组成物, 形成物; 组成, 形成; 编队, 队形 (2)
- *marine** /mə'ri:n/ *adj.* 海的, 海产的; 海船的, 海上贸易的 (2)
- *twilight** /'twɑɪlaɪt/ *n.* 暮色, 黄昏; 没落时期, 衰退期 (2)
- *filter** /'fɪltə(r)/ *n.* 滤器, 过滤器; 滤光器, 滤声器; 筛选 (过滤) 程序 *vt.* 过滤; (用程序) 筛选 (2)
- *evolve** /ɪ'vɒlv/ *vi. & vt.* 进化, 演化; (使) 逐渐形成, 逐渐演变 (2)
- *capture** /'kæptʃə(r)/ *vt.* 俘获, 捕获; 用武力夺取, 攻取; 夺得, 赢得 *n.* (被) 捕获, (被) 俘获 (2)
- *backward-curving** /'bækwəd kɜ:vɪŋ/ *adj.* 向后弯曲的 (2)
- in equal measure** 在同样程度上 (2)
- *solely** /'səʊlli/ *adv.* 仅, 只, 单独地 (3)
- reproduction** /ˌri:prə'dʌkʃn/ *n.* 生殖, 繁殖; 复制; (声音的) 录制; (尤指艺术品的) 复制品 (3)
- *scarce** /skeəs/ *adj.* 缺乏的, 稀少的 (3)
- mate** /meɪt/ *n.* 偶, 伴侣; 朋友, 伙伴 *vi.* 交配 (3)
- *shrimp** /ʃrɪmp/ *n.* (*pl. shrimps or shrimp*) 虾 (3)
- *hydrothermal** /ˌhaɪdrə(u)'θɜ:ml/ *adj.* 热液的, 高温水作用的 (3)
- *vent** /vent/ *n.* (气体、液体的) 出口, 进口 (3)
- *fluid** /'flu:ɪd/ *n.* 液体, 流体 *adj.* 流畅优美的; 易变的, 不稳定的; 流动的 (3)
- *thermal** /'θɜ:ml/ *adj.* 温暖的, 热的; 热量的; 保暖的, 防寒的 (3)
- await** /ə'weɪt/ *vt.* 将发生在, 将降临到 (某人头上); 等候, 等待 (3)
- *submersible** /səb'mɜ:səbl/ *n.* 可潜船, 潜水器 *adj.* 水下使用的 (5)
- *pave** /peɪv/ *vt.* (用砖石) 铺 (地) (5)
- pave the way for** 为...铺平道路, 创造条件 (5)
- *leap** /li:p/ *n.* 跳跃, 跳高; 骤变, 激增 *vt. & vi.* (**leapt, leapt**) 跳, 跳跃; 猛冲, 突然做 (某事); 骤增, 猛涨 (6)
- leap year** 闰年 (6)
- *spin** /spɪn/ *vi. & vt.* (**spun, spun**) (使) 快

速旋转; (使) 急转身; 纺 (线)
n. 高速旋转 (6)

***axis** /'æksɪs/ *n.* (*pl.* **axes** /'æksi:z/) 轴 (旋转物体假想的中心线); 坐标轴; 对称中心线 (6)

***tidal** /'taɪdl/ *adj.* 潮汐的, 有潮的 (6)

lay down 积存; 放下, 停止使用 (6)

***rotation** /rəu'teɪʃn/ *n.* 旋转, 转动; (旋转的) 一周, 一圈; 轮换, 交替 (7)

gravitational /ˌɡrævɪ'teɪʃənl/ *adj.* 引力的, 重力引起的 (7)

***alter** /'ɔ:ltə(r)/ *vt. & vi.* (使) 改变, 更改; 修改 (衣服使更合身) (7)

***pavement** /'peɪvmənt/ *n.* 人行道; 石板铺的地面; 路面 (11)

***crack** /kræk/ *vt. & vi.* 砸开, 砸碎; 破裂, 裂开; 猛击; (使) 发出爆裂声
n. 裂缝; 缝隙, 窄缝; 爆裂声 (11)

***mantle** /'mæntl/ *n.* 地幔; 覆盖层 (11)

as yet 直到现在 (11)

close up (使) 变狭窄; (使) 靠拢, 靠紧; (尤指临时) 关门停业 (11)

***geology** /dʒi'ɒlədʒi/ *n.* 地质; 地质学 (11)

mine /maɪn/ *n.* 矿井, 矿 (11)

***thrive** /θraɪv/ *vi.* 茁壮成长, 蓬勃发展 (11)

enlightening /ɪn'laɪtnɪŋ/ *adj.* 让人受启发的 (11)

***crust** /krʌst/ *n.* 地壳; 面包皮 (11)

innermost /'ɪnəməʊst/ *adj.* 最靠近中心的, 最深处的; 内心深处的 (11)

***fig** /fɪɡ/ *n.* 无花果 (12)

***mirage** /'mɪrɑ:ʒ/ *n.* 蜃景, 海市蜃楼 (13)

***optical** /'ɒptɪkl/ *adj.* 光学的, 视觉的; 有助于视力的 (13)

Unit 2

***plough** /plau/ (*AmE* **plow**) *vt. & vi.* 犁 (田), 耕 (地)
n. 犁 (15)

***fertility** /fə'tɪləti/ *n.* 富饶, 丰产 (15)

***collaboration** /kə'læbə'reɪʃn/ *n.* 合作, 协作; 合作成果 (16)

real time 实时 (16)

put sth to good use 有效使用 (16)

minimize /'mɪnɪmaɪz/ (*BrE* also **minimise**)
vt. 使减少到最低限度; 降低, 贬低; (尤指在计算机屏幕上) 使最小化 (16)

***boast** /bəʊst/ *vt. & vi.* 有 (值得自豪的东西); 自夸, 自吹自擂 (16)

***diagnose** /'daɪəɡnəʊz/ *vi. & vt.* 判断 (问题的原因), 诊断 (疾病) (16)

***moisture** /'mɔɪstʃə(r)/ *n.* 水分, 潮气 (17)

***pH** /ˌpi: 'etʃ/ *n.* pH 值 (溶液或物质的酸碱度) (17)

***fertilizer** /'fɜ:təlaɪzə(r)/ (*BrE* also **fertiliser**)
n. 肥料 (17)

***yield** /jɪld/ *n.* 产量, 产出
vt. 出产 (作物), 产生 (收益等)
vi. 屈服, 让步; 变形, 弯曲 (17)

simplify /'sɪmplɪfaɪ/ *vt.* 使简化, 使简易 (17)

***algorithm** /'ælgərɪðəm/ *n.* 算法, 计算程序 (17)

***spray** /spreɪ/ *vt. & vi.* 喷, 喷洒; 向...扫射 (或抛洒)
n. 浪花, 水花; 喷剂; 喷雾器 (17)

***vertical** /'vɜ:tɪkl/ *adj.* 竖的, 垂直的; 纵向的 (19)

***pesticide** /'pestɪsaɪd/ *n.* 杀虫剂 (19)

***obstacle** /'ɒbstəkl/ *n.* 障碍, 阻碍; 障碍物, 绊脚石 (19)

***silt** /sɪlt/ *n.* 泥沙, 淤泥 (20)

***irrigation** /ˌɪrɪˈɡeɪʃn/ *n.* 灌溉 (20)

***soya bean** /ˈsɔɪbiːn/ (*AmE* **soybean**) *n.* 大豆 (20)

***abundance** /əˈbʌndəns/ *n.* 大量, 充裕 (20)

***Egyptian** /iˈdʒɪpjən/ *adj.* 埃及的, 埃及人的
n. 埃及人 (21)

***flax** /flæks/ *n.* 亚麻; 亚麻纤维 (21)

***drip** /drɪp/ *n.* 水滴, 滴液; 滴落, 滴水声;
(静脉) 滴注
vt. & vi. 滴下; 滴水 (21)

***pest** /pest/ *n.* 害虫; 讨厌的人 (或物) (22)

downside /ˈdaʊnsaɪd/ *n.* 缺点, 不利方面 (22)

***bee** /biː/ *n.* 蜜蜂 (22)

***the pros** /prəʊz/ **and cons** /kɒnz/ 事物的利
与弊 (22)

***conquer** /ˈkɒŋkə(r)/ *vt.* 在 (某地) 很受欢迎;
攻克, 征服; (尤指比赛中) 击败 (25)

***legend** /ˈledʒənd/ *n.* 传说, 传奇故事; 传奇
人物 (25)

nobleman /ˈnəʊblmən/ *n.* (*pl.* **noblemen**)
贵族成员, 出身高贵的人 (25)

***sip** /sɪp/ *vt. & vi.* 小口喝, 抿 (25)

***porter** /ˈpɔːtə(r)/ *n.* 搬运工, 行李员; (医院
里护送病人的) 护工; 门卫, 门房 (25)

***monk** /mʌŋk/ *n.* 僧侣, 修道士 (25)

***Portuguese** /ˌpɔːtʃuˈɡiːz/ *adj.* 葡萄牙的, 葡
萄牙人的, 葡萄牙语的
n. (*pl.* **Portuguese**) 葡萄牙人; 葡萄牙
语 (25)

***Dutch** /dʌtʃ/ *adj.* 荷兰的, 荷兰人的, 荷兰
语的 (25)

***contemplation** /ˌkɒntəmˈpleɪʃn/ *n.* 深思,
沉思; 凝视, 默默注视 (25)

***humid** /ˈhjuːmɪd/ *adj.* 温暖潮湿的 (26)

***well-drained** /ˌwel ˈdreɪnd/ *adj.* 排水良好
的 (26)

***steep** /stiːp/ *adj.* 陡的, 陡峭的; 大起大落
的; 过高的, 不合理的 (26)

***bush** /buʃ/ *n.* 灌木; 灌木状的东西 (26)

maturity /məˈtʃʊərəti/ *n.* (人或动植物) 成
熟, 完全长成; (思想行为等) 成熟 (26)

***bud** /bʌd/ *n.* 芽, 苞; 半开的花, 未长大
的叶 (26)

***exclusion** /ɪkˈskluːʒn/ *n.* 排除在外, 排斥;
排除, 认为不可能 (26)

***oolong** /ˈuːlɒŋ/ *n.* 乌龙茶 (26)

***powdered** /ˈpaʊdəd/ *adj.* 研成粉末的; 干燥
成粉的 (26)

***mint** /mɪnt/ *n.* 薄荷; 薄荷糖 (26)

***cultivation** /ˌkʌltɪˈveɪʃn/ *n.* 耕种, 种植;
(品质或技巧的) 培养 (27)

Unit 3

***casually** /ˈkæʒuəli/ *adv.* 漫不经心地 (30)

***glance** /ɡlɑːns/ *vi.* 瞥一眼; 粗略地看
n. 匆匆一看, 一瞥 (30)

pop-up /ˈpɒp ʌp/ *adj.* 有弹出功能的; (书
籍等) 有立体活动图的
n. (计算机) 弹出式选单或窗口 (30)

***revolution** /ˌrevəˈluːʃn/ *n.* 巨变, 大变革;
革命; (环绕中心点的) 旋转 (30)

hardware /ˈhɑːdweə(r)/ *n.* 硬件 (30)

***radar** /ˈreɪdɑː(r)/ *n.* 雷达 (30)

***lidar** /ˈlaɪdɑː(r)/ *n.* 激光雷达 (30)

***puzzle** /ˈpʌzl/ *n.* 拼图; 谜, 智力游戏; 不解
之谜, 疑问
vt. 迷惑, 使困惑 (30)

***accelerator** /ək'seləreɪtə(r)/ *n.* (汽车等的) 加速装置, 油门; 加速器 (30)

***brake** /breɪk/ *n.* 刹车, 制动器; 阻力, 障碍
vt. & vi. 用闸减速, 刹 (车) (30)

in line with 与...相似 (或紧密相连) (30)

perception /pə'sepʃn/ *n.* 知觉, 感知; 洞察力; 看法, 见解 (30)

ahead of time 提前 (31)

***tragic** /'trædʒɪk/ *adj.* 悲惨的; 悲剧的 (31)

***pedestrian** /pə'destriən/ *n.* 行人, 步行者
adj. 行人使用的, 行人的 (31)

***dilemma** /dɪ'lemə/ *n.* 困境, (进退两难的) 窘境 (31)

***confront** /kən'frʌnt/ *vt.* 使面对; 使...无法回避; 处理, 解决; 面对, 对抗 (31)

be confronted with 面对 (某事物) (31)

***manufacture** /ˌmænju'fæktʃə(r)/ *vt.* 大量生产, 成批制造 (31)

at fault 有过错, 有责任 (31)

regulation /ˌregju'leɪʃn/ *n.* 规章制度, 规则; 管理, 控制 (31)

***cautious** /'kɔːʃəs/ *adj.* 小心的, 谨慎的 (31)

***fare** /feə(r)/ *n.* 车费, 船费 (33)

***cater** /'keɪtə(r)/ *vi. & vt.* (为社交活动) 提供饮食, 承办餐宴 (34)

cater for 满足需要, 适合 (34)

hotspot /'hɒtspɒt/ *n.* 热闹的娱乐场所; 多事之地; 热点 (旅馆、机场等安装设备可无线上网的地方) (34)

upgrade /ˌʌp'ɡreɪd/ *vt.* 提高 (设施等) 档次, 改善; 使 (机器等) 升级; 提升 (34)

***ferry** /'feri/ *n.* 渡船, 轮渡
vt. & vi. 渡运, 摆渡 (34)

***frustration** /frʌ'streɪʃn/ *n.* 懊恼, 沮丧; 令人懊恼 (或沮丧) 的事物; 受挫, 挫败 (35)

***install** /ɪn'stɔːl/ *vt.* 安装, 设置 (35)

***emission** /ɪ'mɪʃn/ *n.* 排放物, 散发物; (光、热、气等的) 发出, 排放 (35)

***distracting** /dɪ'stræktɪŋ/ *adj.* 令人分心的 (36)

***blink** /blɪŋk/ *n.* 眨眼睛
vt. & vi. 眨眼睛; 闪烁 (36)

in the blink of an eye 眨眼的工夫, 很快 (36)

likelihood /'laɪklihood/ *n.* 可能, 可能性 (36)

***alert** /ə'leɪt/ *adj.* 警觉的; 意识到 (问题或危险)
vt. 使警觉; 使意识到 (36)

severity /sɪ'verəti/ *n.* 严重性; 严厉 (36)

***bullet-nosed** /ˌbʊlɪt 'nəʊzɪd/ *adj.* 子弹头的 (39)

***glide** /ɡlaɪd/ *vi.* 滑行, 掠过; 滑翔
n. 滑行, 滑翔 (39)

***hiss** /hɪs/ *n.* 嘶嘶声, 嘘声
vi. & vt. 发嘶嘶声; 发嘘声 (39)

spacious /'speɪʃəs/ *adj.* 宽敞的 (39)

***swiftly** /'swɪftli/ *adv.* 迅速地; 敏捷地 (39)

in operation 使用中, 工作中 (39)

blueprint /'bluːprɪnt/ *n.* 计划蓝图, 行动方案; (建筑、机器等的) 蓝图 (39)

***horizontal** /ˌhɒrɪ'zɒntl/ *adj.* 横的, 水平的 (39)

visionary /'vɪʒənəri/ *adj.* 有远见卓识的 (39)

***perseverance** /ˌpɜːsə'veɪərəns/ *n.* 毅力, 不屈不挠的精神 (40)

***debut** /'deɪbjʊː/ *n.* 初次登台 (或上场); (演员、运动员的) 首次亮相 (40)

make one's debut 首次亮相 (40)

***revive** /rɪ'vaɪv/ *vt. & vi.* 重新使用; (使) 苏醒, 复活 (40)

***sector** /'sektə(r)/ *n.* 部门, 领域; 地带 (40)

***nightmare** /'naɪtmɛə(r)/ *n.* 可怕的经历; 噩梦 (41)

Unit 4

***twinkle** /'twɪŋkl/ *vi.* 闪烁; (眼睛因兴奋)

发亮

n. 闪耀, 闪烁; (眼睛的) 闪亮 (43)

***steady** /'stedɪ/ *adj.* 稳定的, 恒定的; 稳步

的, 匀速的; 平稳的, 稳固的 (43)

***density** /'densəti/ *n.* 密度; 密集, 稠密 (43)

***horizon** /hə'reɪzn/ *n.* 地平线, 海平线;

(知识或兴趣的) 范围, 眼界 (43)

***fragrant** /'freɪgrənt/ *adj.* 香的, 芳香的 (43)

***pollen** /'pɒlən/ *n.* 花粉 (43)

***microbiome** /ˌmaɪkrəʊ'baɪəʊm/ *n.* 微生物

组, 微生物群系 (44)

***naked** /'neɪkɪd/ *adj.* 裸露的; 无遮盖的 (44)

the naked eye 肉眼 (44)

***trillion** /'trɪljən/ *n.* 万亿, 兆; 大量, 无数 (44)

***blend** /blend/ *n.* 混合品, 混合物; 融合

vt. 使混合

vi. (和某物) 融合 (44)

foothold /'fʊθəʊld/ *n.* 稳固地位, 立足点;

立足处 (44)

***gut** /gʌt/ *n.* 肠道, 消化道; (*pl.*) (尤指动物

的) 内脏 (44)

***obesity** /əʊ'bi:səti/ *n.* 肥胖 (45)

***cognitive** /'kɒɡnətɪv/ *adj.* 认知的 (45)

bloodstream /'blʌdstri:m/ *n.* 体内循环的

血液, 血流 (45)

nerve /nɜ:v/ *n.* 神经; (*pl.*) 神经质, 神经

紧张; 勇气, 气魄 (45)

transplant /'trænsplɑ:nt/ *n.* (器官等的) 移

植; 移植器官

/træns'plɑ:nt/ *vt.* 移植 (器官) (45)

***vaccine** /'væksɪn/ *n.* 疫苗 (45)

***malnutrition** /ˌmælnju'trɪʃn/ *n.* 营养不良 (45)

***droplet** /'drɒplət/ *n.* 小滴 (47)

***isolation** /ˌaɪsə'leɪʃn/ *n.* 隔离, 隔离状态;

孤独, 孤立状态 (47)

***inject** /ɪn'dʒekt/ *vt.* (给...) 注射 (药物等);

(给...) 添加, 增加 (47)

***sequence** /'si:kwəns/ *vt.* 测定 (整套基因或

分子成分的) 序列; 按顺序排列

n. 一系列, 一连串; 顺序, 次序 (47)

***snowflake** /'snəʊfleɪk/ *n.* 雪花, 雪片 (48)

alike /ə'laɪk/ *adj.* 相像, 十分相似

adv. 十分相像地; 两者都, 同样地 (48)

***crystal** /'krɪstl/ *n.* 结晶, 晶体; 水晶 (48)

***melt** /melt/ *vi. & vt.* 融化, (使) 熔化; (使)

软化, 变得温柔 (48)

***feather** /'feðə(r)/ *n.* 羽毛, 翎毛 (48)

***miracle** /'mɪrəkl/ *n.* 极好的例子, 精品;

奇迹, 不平凡的事 (48)

***particle** /'pɑ:tɪkl/ *n.* 颗粒, 微粒; 粒子 (49)

***vapour** /'veɪpə(r)/ (*AmE vapor*) *n.* 蒸气,

雾气 (49)

***molecule** /'mɒlɪkjʊ:l/ *n.* 分子 (49)

***accumulate** /ə'kju:mjəleɪt/ *vt.* 积累, 积聚

vi. (数量) 逐渐增加 (49)

***humidity** /hju:'mɪdəti/ *n.* 湿度; 高温潮湿 (49)

resist /rɪ'zɪst/ *vt. & vi.* 忍住, 抵挡; 抵制, 阻

挡; 反抗, 抵抗 (50)

***texture** /'tekstʃə(r)/ *n.* 质地, 手感; 口感 (50)

***lubrication** /ˌlu:bri'keɪʃn/ *n.* 润滑, 润滑

作用 (50)

***bionics** /baɪ'ɒnɪks/ *n.* 仿生学 (53)

***intersect** /ˌɪntə'sekt/ *vi. & vt.* 相交, 交叉;

横穿, 贯穿 (53)

***carpenter** /'kɑ:pəntə(r)/ *n.* 木工, 木匠 (53)

***saw** /sɔ:/ *n.* 锯

vt. & vi. (**sawed, sawn**) 锯; 拉锯似的

来回移动 (53)

fit in with 适应, 与...合得来 (53)

***spiral** /ˈspaɪrəl/ *adj.* 螺旋形的,螺旋式的
 n. 螺旋形,螺旋式 (53)
cancel out 抵消,对消 (54)
intensify /ɪnˈtensɪfaɪ/ *vt. & vi.* (使) 加强,
 增强 (54)
be inset into 被嵌入 (54)
banknote /ˈbæŋknəʊt/ *n.* 纸币 (54)

***exoskeleton** /ˈeksəʊskelɪtn/ *n.* (昆虫等动
 物的) 外骨骼 (54)
backdrop /ˈbækdɒp/ *n.* 周围陪衬景物;
 (舞台的) 背景幕布 (54)
***hook** /hʊk/ *n.* 钩,挂钩;诱饵,吸引人的
 事物 (55)
***cube** /kjuːb/ *n.* 立方形的东西;立方体 (55)

